H10 and D2
Sleep comparison

A 20-page analysis of synchronized overnight sleep and naps
Real recordings only | 4-6 November 2026 | Singapore time (UTC+8)

Main finding: sleep duration is similar, but stage timing is not interchangeable. Across the two nights, stage labels match for **44.9%** of shared minutes; chance-adjusted agreement is **-0.013**.

What this report establishes

Two overnight pairs and two nap pairs were independently checked against the exported timelines. Overnight total-sleep differences are 10.5 and 3.5 minutes, with D2 reporting less sleep in both. Agreement on sleep versus wake is 94.8% across the nights, but most shared minutes are sleep, so that percentage does not demonstrate reliable wake detection.

The D2 naps contain code 5: **asleep, without a stage**. OpenBeat includes that code in its displayed light-sleep total, although its comparison engine correctly excludes it from stage agreement. It would be misleading to conclude that the D2 measured 100% light sleep during these naps.

What the devices actually do

The H10 supplies electrical beat timing and chest acceleration; OpenBeat assigns its sleep stages using a transparent heuristic. D2 supplies sleep labels already calculated by Huawei firmware; OpenBeat imports and compares them. Better beat timing does not by itself establish better sleep staging.

Evidence base	Scope
Paired observations	2 nights + 2 naps; one person
Shared comparison time	657 overnight minutes; 152 nap minutes
Software	OpenBeat 1.32.1; saved sleep method 2
Reference standard	No EEG / EOG / chin EMG or polysomnography

Prepared 6 November 2026. The user-requested demo exclusion is applied throughout. No demo results are analyzed, plotted or pooled. This is a device/software comparison, not a clinical sleep diagnosis.

01 Evidence and scope

Which records are eligible, and what is deliberately outside the comparison.

The backup was exported on 6 November 2026 at 18:57:50 Singapore time. Eligible paired results are taken from the current, non-demo sleepReports entries and their embedded polarCopy and watchCopy snapshots. Each is a saved view of one simultaneous recording, not a new independent night every time a report is regenerated.

Record	H10 recording window (SGT)	D2 sleep window	Shared
Nap 1 - 4 Nov	10:57:58-11:54:32	11:07-11:51	44 min
Night 1 - 4/5 Nov	23:31:01-06:30:08	23:56-06:28	392 min
Night 2 - 6 Nov	01:53:09-06:32:12	02:03-06:28	265 min
Nap 2 - 6 Nov	15:56:08-17:52:42	16:04-17:52	108 min

Two additional H10 sessions are unpaired

On 4 November, H10 recordings beginning at 07:40:52 (109.3 minutes) and 10:05:20 (13.3 minutes) have no temporally overlapping D2 sleep entry in the supplied backup. They are listed for completeness but are excluded from paired device differences. Missing a paired entry here does not prove the watch failed to record sleep; the necessary record may not have been exported or the wearer may have been awake.

Data layers used

Numerical evidence: saved 30-second H10 stages and feature arrays, D2 minute codes and summary values, synchronized minute arrays and sleep summaries. **Implementation evidence:** the supplied TypeScript source and changelog. **Physiology:** manufacturer documentation and clinical sleep references, distinguished from this person's observations.

Earlier report versions in past are not treated as extra observations. Other ECG sessions, blood-pressure downloads, older watch-only nights and device-discovery data do not enlarge the synchronized sample. All conclusions about agreement apply to these four eligible pairs.

Sources: backup sleep, sleepReports and d2.nights; supplied src/syncsleep.ts and src/d2/night.ts. All clock times in this report are Asia/Singapore.

02 How the comparison was calculated

Time alignment, denominators and independent checks.

OpenBeat pairs sessions when overlap is at least half the shorter recording, chooses the largest overlap first and uses each recording once. This is temporal matching, not evidence that the devices share a hardware clock or a common sleep-stage model.

Step	Verified implementation / report treatment
H10 epochs	One label per 30 seconds. Epoch start timestamps are assigned to clock-minute slots.
Minute conversion	Most common H10 label wins; a tie retains the first label encountered. This is not fractional-duration weighting.
D2 input	One imported code per minute; displayed sleep is trimmed to fallAsleepTime through wakeupTime.
Shared minutes	Only slots containing both devices are compared. Missing slots remain missing.
Stage agreement	Same label / minutes with specific labels on both devices. Generic asleep is excluded.
Binary agreement	Same asleep-or-awake side / all shared minutes, including generic asleep.
Chance adjustment	Cohen's kappa uses the four-stage row and column totals. Neither device is ground truth.

Independent reconciliation

For all four pairs, I reconstructed the saved minute arrays from the embedded H10 stage strings, session starts and raw D2 stage lists. The reconstructed arrays match the exported synchronized arrays. I also recomputed shared time, stage agreement, sleep/wake agreement and kappa; all match the stored values to numerical precision.

Full-recording sleep totals retain the app's original 30-second precision. Agreement uses minute labels only over overlap. Therefore stage totals in a full-session summary need not equal row totals in a shared-minute confusion matrix. H10 awakeMin is wake after sleep onset within its detected sleep period; it excludes pre-onset and post-period awake time.

Source: supplied src/syncsleep.ts: pairUp, strapMinutes, lineUp, compareMinutes; src/d2/night.ts: figures. The lag experiment on page 17 is an additional analysis, not an alteration of the saved data.

03 What sleep stages mean

Clinical definitions versus the indirect evidence available to wearables.

Label	Clinical basis	What these wearables can infer
Awake	Brain activity, eye activity and behavior support wakefulness.	Movement, posture and cardiac changes; quiet wake can resemble sleep.
Light (N1/N2)	N1 is transitional; N2 includes characteristic EEG spindles and K-complexes.	A sleep-like period without enough evidence for the model's deep or REM rule.
Deep (N3)	Slow-wave activity identified from EEG.	Relatively stable cardiorespiratory patterns and stillness can be suggestive, not confirmatory.
REM	Characteristic brain/eye activity with low chin muscle tone.	Low movement with changing heart/respiratory patterns can suggest REM; the eyes and chin are not measured.

Why cardiac sensors cannot directly see a stage

Clinical staging uses signals that distinguish brain state, eye movements and muscle tone. Neither an H10 chest ECG nor a wrist pulse sensor records the defining brain-wave patterns. Their stage labels are predictions from correlated body signals. The absence of movement is especially ambiguous: a person can lie awake almost motionless.

Deep sleep tends to be concentrated earlier in a typical night, with more REM later. Such patterns can improve a statistical estimate but are not requirements for every nap or every individual. They should not be mistaken for measurements of brain activity. [4]

What disagreement means here

When one system says deep and the other says light, the data establish different labels, not which one is right. Similar totals can arise from episodes placed at different times. Conversely, a close match could reflect shared assumptions rather than clinical accuracy. Validation requires simultaneous reference sleep scoring, ideally with the relevant device and firmware version. [5]

Interpretation rule: use duration and timing as useful estimates; treat deep/REM percentages and exact stage transitions as more uncertain. Do not diagnose insufficient deep sleep, REM abnormalities or a sleep disorder from these charts alone.

References [4]-[5] are listed on page 20. The clinical-stage summary is distinct from the actual OpenBeat rules described on pages 6-7.

04 H10: measurements and processing

The chest strap measures signals; OpenBeat builds the sleep report.

Polar's SDK documents H10 heart rate and RR intervals, ECG at 130 Hz and acceleration at selectable rates. The supplied sleep records use acceleration at 25 Hz. The OpenBeat sleep classifier uses **RR intervals plus chest acceleration**; it does not classify stages from QRS duration, QTc, ST segments or ECG waveform morphology. [1, S1]

Signal / feature	What the supplied implementation does
Beat intervals	Accepts 300-2000 ms intervals within 20% of a local neighbor median; rejected intervals do not contribute to the cardiac features.
Epoch heart rate	60000 / mean accepted RR, requiring at least 10 accepted beats in the 30-second epoch.
RMSSD	Successive clean RR differences pooled over approximately 5 minutes; at least 60 differences required. Gaps are not bridged.
Heart-rate spread	Standard deviation of available epoch heart rates across 5 epochs (2.5 minutes), requiring at least 3 values.
Movement / posture	Acceleration magnitude changes and the mean gravity direction describe chest movement and turning.
Respiration proxy	Chest acceleration is filtered in a respiratory band; beat-interval modulation supplies another regularity estimate.

Breathing is estimated, not airflow measured

The code selects the chest axis with greatest respiratory-band energy in a roughly 60-second window. It requires at least 40 seconds of data, 90% valid samples, enough cycles and a rate between 5 and 35 breaths/min. A similarly filtered RR series contributes breath-to-breath regularity. This does not measure airflow, oxygen desaturation or obstructive events. [S1]

This processing explains why an acceleration feature can remain available while heart rate or breathing is missing. Signal cleaning reduces artefact contamination, but may also remove genuine irregular beats; a sleep stage assigned during such periods is not an independent rhythm assessment.

Sources: [1] Polar SDK; [S1] src/sleep.ts: heartFeatures, heartBreathing, accFeatures, breathsIn. RMSSD values are OpenBeat estimates, not a demonstrated equivalent of Huawei's HRV summary.

05 H10: asleep versus awake

The actual detection rules are more specific than “low heart rate means sleep”.

Rule	Implementation in sleep method 2
Movement threshold	max(8 mG, 2.5 x median epoch activity). Strong movement is a wake candidate.
Turning	A greater-than-20-degree change in mean chest orientation is a wake candidate.
Heart rate + movement	HR above the 20th-percentile HR baseline by more than 12 bpm, together with activity above half the movement threshold, is a wake candidate.
Temporal filter	At least 2 of 3 neighboring epochs must be wake candidates. A single isolated turn does not necessarily become wake.
Detected onset	First start of a continuous 20-epoch (10-minute) run that has data and is not wake.
Detected end	End of the last qualifying 10-epoch (5-minute) sleep-like run, with a fallback if necessary.
No-data condition	An epoch is marked no-data only when both activity and epoch HR are unavailable.

Practical consequences

Onset is backdated to the start of the qualifying quiet run. It is not an observed EEG transition and cannot be known at that instant without later data. A person who lies quietly while awake can satisfy the rule. Elevated HR alone does not trigger the combined HR rule if movement is insufficient.

The beginning and end of a recording matter. Starting after falling asleep can make latency appear zero. The final reported wake may sit close to the recording end because a last quiet run qualifies as sleep; that does not establish the exact moment of waking. These effects are relevant to the 6 November nap.

Why awakening counts differ

H10 awakenings count only awake runs of at least 2 minutes inside the detected sleep period. The D2 fallback in src/d2/night.ts counts each awake-code run between sleep, without that two-minute minimum. Thus the overnight counts of 1 versus 5 and 0 versus 2 are not directly equivalent measures of fragmentation.

Source: [S1] analyzeSleep; [S3] countWakeups. Both stages and timing are algorithmic estimates; “has data” does not mean all staging inputs are present.

06 H10: light, deep and REM rules

The transparent classifier, including its timing assumptions.

OpenBeat calculates relative features within the detected sleep period. HR, RMSSD and HR spread are expressed as deviations from nearby sleep, using up to 45 minutes on either side and scaling by residual standard deviation. Respiratory irregularity is standardized across the sleep mask and combines chest and RR-derived estimates when both exist. [S1]

Deep score = -zHR - zBreathingCV + 0.5 zRMSSD - 0.5 zHRspread + 0.5(1-2t)

REM score = zHR + zHRspread + zBreathingCV - 0.3 zRMSSD + 0.5(2t-1)

t is progress through a sleep period of at least 3 hours; for shorter sleep t=0.5, making the time-prior terms zero.

Decision order

1. If deep score exceeds 1.2, label deep. **2.** Otherwise, if REM score exceeds 1.2 and elapsed time since onset is greater than 50 minutes, label REM. **3.** Otherwise label light. Missing normalized feature terms are substituted with zero. Awake and no-data checks occur before these stage decisions.

The scores are heuristic evidence, not calibrated probabilities. A value of 1.2 does not mean 120%, and the coefficients do not demonstrate validated staging accuracy. Deep takes priority if both scores would cross threshold. A locally lower HR and more regular breathing favor deep; relatively variable HR/breathing favor REM.

Smoothing changes the final labels

A 7-epoch mode window reduces flicker across about 3.5 minutes. A minimum-bout pass joins some brief gaps and converts short deep/REM bouts to light, using a nominal 5-minute minimum. Awake/no-data epochs remain protected. These rules can suppress brief episodes even if their feature scores initially cross threshold.

Nap implication: REM is not directly eligible during the first 50 minutes after onset, leaving very little eligibility in the short nap. The longer nap has no early/late time prior because its detected period is under 3 hours. This is a design constraint, not proof that early REM is biologically impossible.

Source: [S1] localZ, analyzeSleep, minBouts. Changelog v1.29.0 introduced method 2; all four current paired reports carry method=2.

07 D2: what is known and unknown

Huawei stages are imported; the supplied app does not recreate TruSleep.

The D2 has an optical heart-rate sensor and an inertial measurement unit, alongside other health sensors. Huawei describes TruSleep as deriving sleep states from motion and physiological features. The exact model weights, decision thresholds and firmware-specific processing are not available in this backup or the supplied OpenBeat source. [2,3]

Layer	What can be verified
Watch hardware	Wrist optical pulse sensing and motion sensing are relevant to overnight tracking. An ECG sensor being present does not establish continuous sleep ECG use.
Huawei processing	Sleep labels and summaries arrive already computed. The backup does not identify the active TruSleep version; sleepVersion is null.
Imported stage codes	1 = light; 2 = REM; 3 = deep; 4 = awake; 5 = generic nap sleep.
OpenBeat comparison	watchMinute maps 5 to asleep, preserving “no specific stage” for agreement.
OpenBeat summary	D2 figures adds code 5 to lightMin for display totals. That aggregation should not be interpreted as measured N1/N2.

The overnight pipeline

In these records, the D2 provides minute-level labels plus average HR, HRV, respiration and oxygen saturation. OpenBeat trims the reported sleep chart to the watch's fallAsleepTime and wakeupTime. Its “bedMs” field is therefore sleep onset, although the name can sound like time entering bed. Actual bedTime is a separate watch summary field. [S3]

The nap pipeline

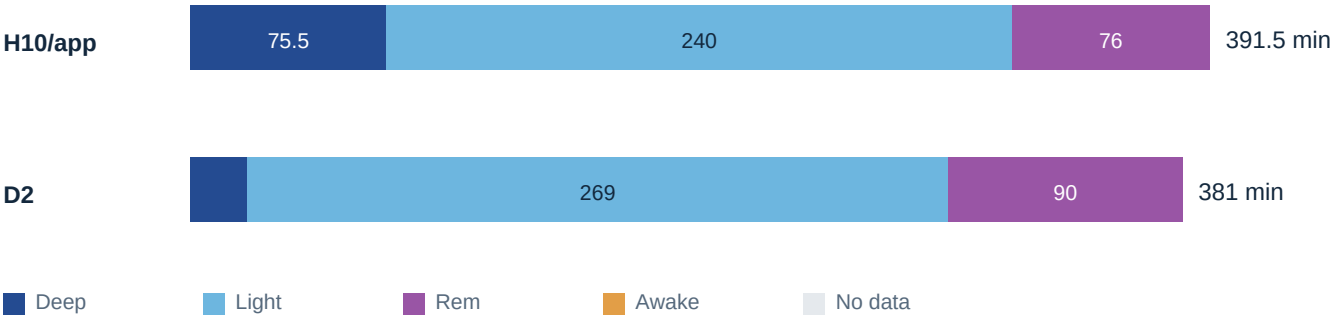
Every D2 minute in both paired naps is code 5. There is no D2 light/deep/REM sequence to validate against the H10 nap sequence. Its zero HR, HRV, breathing and SpO2 summaries are unavailable placeholders in context, not physiological values. Their proper treatment is N/A, with no numerical difference or fit warning generated from them.

Newer Huawei general documentation discusses several TruSleep generations, but that does not establish which one ran on this D2. This report therefore explains the documented sensor principles without assigning an unverified version or claiming access to Huawei's proprietary staging model.

Sources: [2]-[3], [6]; [S2] watchMinute; [S3] figures and isNap. The code-5 finding is directly observed in both watchCopy stage arrays.

08 Night 1: sleep quantities

4 November into 5 November | H10 recording 23:31-06:30.



Measure	H10 + OpenBeat	D2	D2 minus H10
Total sleep	391.5 min	381 min	-10.5 min
Deep	75.5 min	22 min	-53.5 min
Light	240 min	269 min	+29 min
REM	76 min	90 min	+14 min
Awake within sleep window	15 min	11 min	Different boundaries
Awakenings	1	5	Different definitions
Sleep score	95	71	Different scoring systems

Interpretation

Total sleep differs by only 10.5 minutes, but the allocation differs substantially. OpenBeat assigns 75.5 minutes of deep sleep versus D2's 22 minutes: a 53.5-minute difference. Deep sleep is 19.3% versus 5.8% of each device's own sleep total. D2 assigns more light and REM time.

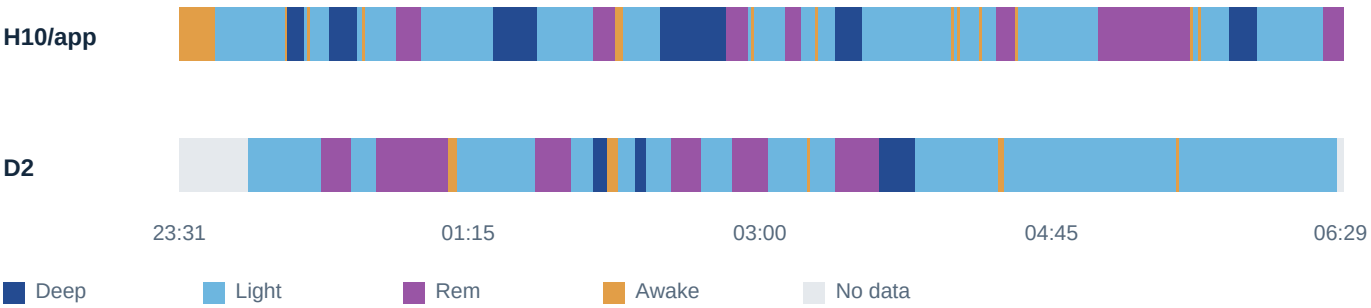
OpenBeat detects onset at 23:43:31; D2 at 23:56:00, 12.47 minutes later. D2 ends sleep about 2.03 minutes earlier. These boundary differences explain part of the total-sleep gap, but cannot explain the large disagreement within their shared interval.

The score difference, 95 versus 71, is not an accuracy test. OpenBeat applies its own duration/efficiency/stage penalties; Huawei's score is imported. A high score can coexist with low device-to-device stage agreement. The next page evaluates whether the deep and REM episodes occur at the same times.

Sources: eligible report n1793806261560+d2-1793807280. Bars show full-session sleep minutes, excluding awake. H10 awakeMin counts internal wake only; awakenings and scores use different rules.

09 Night 1: stage timing

Same-stage agreement 45.2% | sleep/wake agreement 94.1% | kappa -0.001.



H10 row / D2 column	Deep	Light	REM	Awake
Deep	0	46	30	0
Light	17	161	44	4
Rem	5	50	15	6
Awake	0	12	1	1

Confusion matrix: minutes, H10/OpenBeat rows and D2 columns; shared interval only. Diagonal entries are matching labels.

The deepest disagreement is temporal

Across 392 shared minutes, 177 stage labels match. Of the 76 shared minutes labelled deep by H10/OpenBeat, **none** are labelled deep by D2: 46 are light and 30 are REM on D2. Conversely, D2's 22 deep minutes occur when OpenBeat says light (17) or REM (5).

Only 15 shared minutes are REM on both systems. Another 94 minutes swap light and REM in one direction or the other. Five-minute main-stage agreement is 48.7%, so modest smoothing does not resolve the discrepancy.

Why the binary percentage looks much better

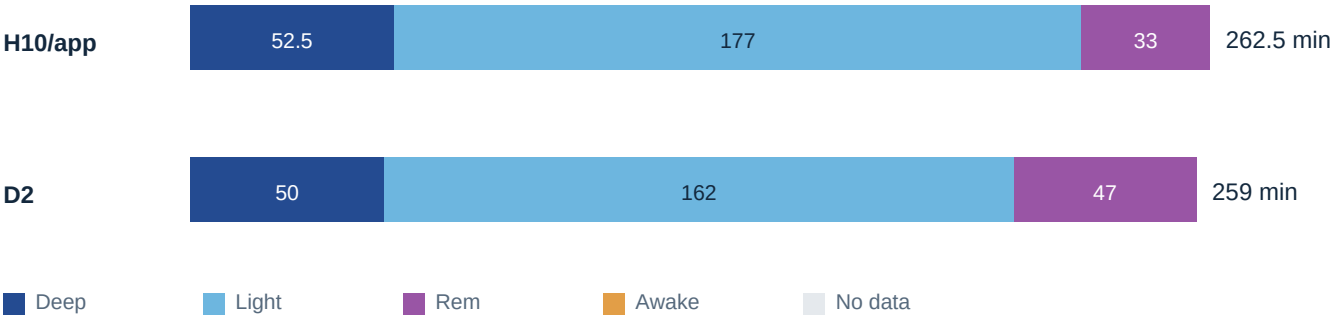
Both systems say asleep for 368 shared minutes and awake for only 1 minute. They disagree on sleep/wake for 23 minutes. Most of the 94.1% agreement therefore comes from the dominant asleep class. Only 1 of 14 H10-labelled awake minutes is also awake on D2 within overlap; this is a relative match rate, not sensitivity against a clinical reference.

Night 1 conclusion: duration agreement is useful, but the deep-sleep totals and episode placements are not interchangeable. Neither the matrix nor the H10's electrical measurement tells us which stage sequence is clinically correct.

Sources: saved minute arrays independently reconstructed and recomputed. Timeline colors are the exported labels at the original clock alignment; grey denotes no D2 data.

10 Night 2: sleep quantities

6 November | H10 recording 01:53-06:32.



Measure	H10 + OpenBeat	D2	D2 minus H10
Total sleep	262.5 min	259 min	-3.5 min
Deep	52.5 min	50 min	-2.5 min
Light	177 min	162 min	-15 min
REM	33 min	47 min	+14 min
Awake within sleep window	5 min	6 min	Different boundaries
Awakenings	0	2	Different definitions
Sleep score	71	63	Different scoring systems

A similar total can conceal different episodes

The two systems estimate 262.5 and 259 minutes asleep, a difference of just 3.5 minutes. Deep totals also look similar: 52.5 versus 50 minutes, about 20.0% versus 19.3% of sleep. That apparent closeness is not evidence that they identify the same deep episodes.

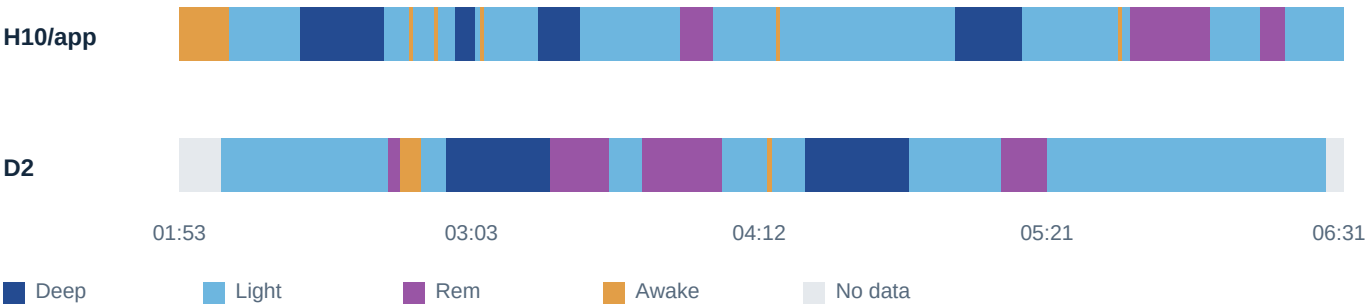
OpenBeat assigns 33 minutes of REM (12.6%); D2 assigns 47 minutes (18.1%). Light totals are 177 versus 162 minutes. The 14-minute REM excess on D2 happens to be identical to Night 1, but two observations are insufficient to establish a stable device bias.

OpenBeat onset is 02:04:39; D2 onset is 02:03:00, about 1.66 minutes earlier. D2 wake is 06:28:00 versus 06:32:09 from OpenBeat, about 4.16 minutes earlier. Because this is only a partial overnight opportunity, it should not be treated as a full-day total or proof of habitual short sleep.

Source: eligible report n1793901189844+d2-1793901420. Full-session totals preserve 30-second H10 precision. The score 71 versus 63 reflects different systems; it is not a probability of healthy sleep.

11 Night 2: stage timing

Same-stage agreement 44.5% | sleep/wake agreement 95.8% | kappa -0.028.



H10 row / D2 column	Deep	Light	REM	Awake
Deep	8	31	12	0
Light	41	101	27	5
Rem	0	25	8	0
Awake	1	5	0	1

Confusion matrix: shared minutes; rows H10/OpenBeat, columns D2. Full-session totals on page 11 and minute counts here use different denominators.

Deep totals agree; deep timestamps mostly do not

Of 51 shared H10-deep minutes, only 8 are also deep on D2. The remaining 43 are light (31) or REM (12) on D2. Of D2's 50 deep minutes, 41 occur during H10-light and 1 during H10-awake. This demonstrates why comparing stage percentages alone is insufficient.

Only 8 of the 33 H10-REM minutes match D2-REM; the other 25 are D2-light. A further 27 minutes are H10-light and D2-REM. Combined, light/REM exchanges account for 52 shared minutes.

Binary matching remains dominated by sleep

Both devices label 253 minutes asleep and 1 minute awake; 11 minutes disagree on sleep/wake. The shared four-stage agreement is 118/265, or 44.5%. Five-minute block agreement reaches only 47.2%. Neither measure provides evidence that exact stage boundaries are dependable between these systems.

Night 2 conclusion: matching total sleep and nearly matching deep-sleep amounts can coexist with poor agreement on when deep sleep occurs. The two reports describe different inferred sleep architecture.

Sources: eligible current saved report and reconstructed minute arrays. Kappa below zero means less agreement than expected from these observed label frequencies, not that either device has negative clinical accuracy.

12 What the two nights establish

Pooled descriptive agreement over 657 shared minutes.

Metric	Night 1	Night 2	Pooled nights
Shared minutes	392	265	657
Exact stage matches	177	118	295
Same-stage agreement	45.15%	44.53%	44.90%
Kappa	-0.0009	-0.0280	-0.0129
Sleep/wake matches	369	254	623
Sleep/wake agreement	94.13%	95.85%	94.82%

H10 row / D2 column	Deep	Light	REM	Awake
Deep	8	77	42	0
Light	58	262	71	9
Rem	5	75	23	6
Awake	1	17	1	2

The strongest quantitative conclusion

Observed stage agreement is 44.90%; agreement expected from the pooled stage frequencies is 45.60%. Hence pooled kappa is approximately -0.013. The result supports **poor interchangeability of stage labels** in this small sample. It does not establish which algorithm is accurate against EEG.

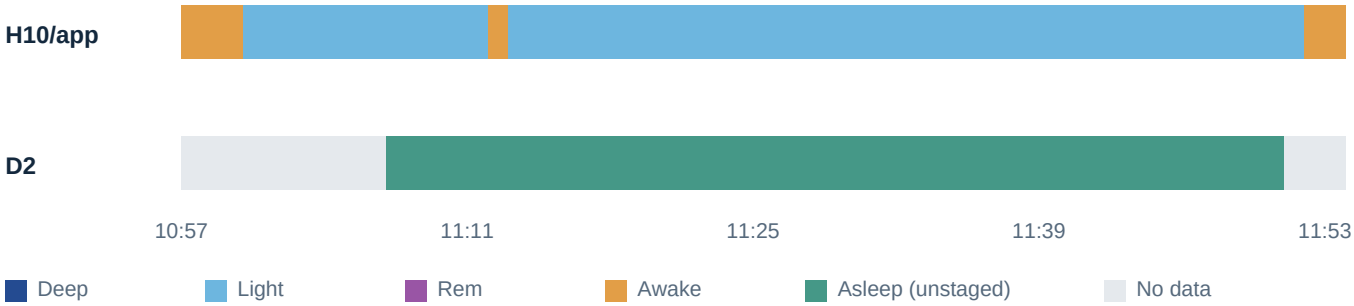
Among H10-labelled minutes, the same D2 label occurs in 8/127 deep minutes (6.3%), 262/400 light minutes (65.5%), 23/109 REM minutes (21.1%) and 2/21 awake minutes (9.5%). These are directional match rates relative to H10 labels, not clinical sensitivity or specificity.

An always-asleep label would match D2's asleep/awake labels in 640/657 minutes (97.4%), exceeding the observed 94.8% binary agreement. That simple prevalence check shows why the high sleep/wake percentage should not be taken as a validated wake detector.

Minute-weighted pooling is descriptive; epochs are autocorrelated and there are only two nights from one person. No population confidence interval or statistical superiority claim is warranted. Mean full-session D2 minus H10 sleep difference is -7.0 minutes.

13 Nap 1: 4 November

H10 recording 10:57:58-11:54:32 | D2 nap 11:07-11:51.



Measure	H10 + OpenBeat	D2
Total sleep	51.5 min	44 min
Specific sleep stages	51.5 min light; 0 deep; 0 REM	Not supplied: all 44 codes are 5
Shared minutes	44	44
Sleep/wake agreement	43/44 = 97.7%	Binary comparison only
Stage agreement / kappa	N/A	No stage-labelled D2 minutes
Average HR / HRV	64.9 bpm / 50.5 ms	N/A; exported zeros
Breathing rate	11.7 breaths/min	N/A; exported zero

What can be compared

D2 begins the nap about 7.02 minutes after the H10 onset estimate and ends it about 1.48 minutes earlier. Full-session estimated sleep differs by 7.5 minutes. Because D2 covers a narrower interval, this is not a pure classifier difference over identical recording time.

Within the 44 shared minutes, H10 agrees with the generic D2 sleep indication for 43 minutes. This is consistent with both detecting a quiet sleep-like period, but cannot validate light sleep. D2 provides no staged reference during this nap.

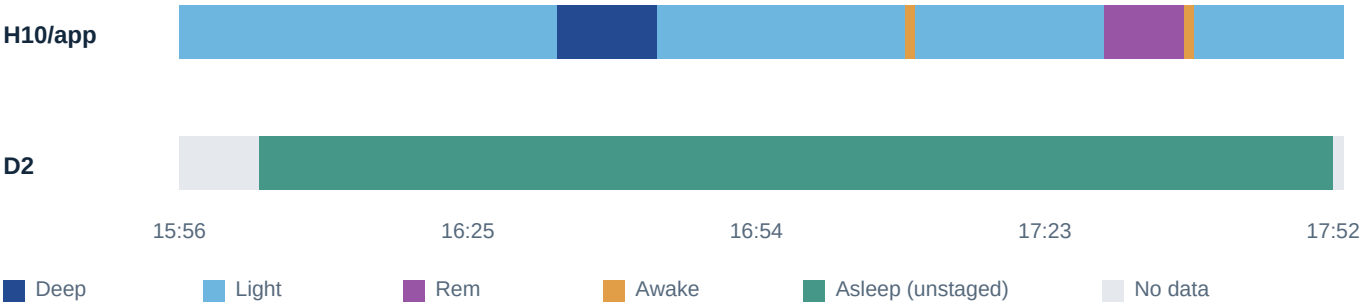
How OpenBeat's rules shape this result

The H10 sleep period spans about 52.5 minutes including its internal wake. REM is only eligible after more than 50 minutes since onset, and brief REM runs are subject to the bout rule. Its absence is therefore partly constrained by the algorithm. The zero deep estimate means its deep rule did not survive final classification; it does not prove no N3 occurred.

Source: report n1793761078822+d2-1793761620; D2 code 5 is unstaged sleep. The exported suggestion about a 65 bpm HR gap is invalid because it subtracts a missing D2 value represented as zero.

14 Nap 2: 6 November

H10 recording 15:56:08-17:52:42 | D2 nap 16:04-17:52.



Measure	H10 + OpenBeat	D2
Total sleep	114.5 min	108 min
Light / deep / REM	95.5 / 10.5 / 8.5 min	Specific stages unavailable
Shared minutes	108	108
Sleep/wake agreement	106/108 = 98.1%	Binary comparison only
Stage agreement / kappa	N/A	Generic asleep code only
Average HR / HRV	60.7 bpm / 35.7 ms	N/A; exported zeros
Breathing rate	12.6 breaths/min	N/A; exported zero

This is not a D2-versus-H10 deep-sleep disagreement

OpenBeat labels deep and REM episodes in this longer nap. D2 does not provide those categories in the exported nap sequence. Its zero deep and REM totals are a consequence of generic code-5 data, not opposing measurements showing those stages were absent.

D2 begins 7.86 minutes after the H10 onset estimate and ends only 0.64 minutes earlier. H10 latency is zero because the first qualifying quiet run starts at the recording beginning. The app does not know whether sleep actually began at that moment or before recording began.

Pooled nap result

Across both naps, 149/152 shared minutes agree on sleep versus wake (98.0%). All shared D2 nap minutes are generic sleep, so this statistic simply counts how often H10 also labels sleep during D2's selected nap interval. It does not test D2's ability to identify awake periods or distinguish stages.

The longer nap retains a saved H10 report but is cloudOnly in the backup and has no raw sleep signal payload there. Its stage sequence and feature completeness can be audited, but the underlying RR/acceleration classifier cannot be rerun from that payload alone.

Source: report n1793951768376+d2-1793952240. Do not interpret stored nap score 47 as a current nap-quality rating; the UI policy suppresses nap scoring.

15 Physiology and missing features

Averages do not compare identical sensors, windows or HRV definitions.

Recording	HR H10 / D2 bpm	HRV H10 / D2 ms	Respiration H10 / D2 breaths/min
Night 1	51.9 / 50	45.3 / 41	12.2 / 14
Night 2	53.9 / 61	50.7 / 48	11.6 / 13
Nap 1	64.9 / N/A	50.5 / N/A	11.7 / N/A
Nap 2	60.7 / N/A	35.7 / N/A	12.6 / N/A

Cardiac differences deserve quality checks, not automatic blame

D2 minus H10 mean HR is -1.9 bpm on Night 1 and +7.1 bpm on Night 2. Mean HRV differs by -4.3 and -2.7 ms. These are full-summary differences, not synchronized beat-by-beat errors. Huawei's HRV definition and averaging windows are not established by these fields; H10 avgHrv is OpenBeat's average of windowed RMSSD while asleep.

Missing H10 feature epochs	HR	Breathing	Activity
Night 1 (838 epochs)	51 (6.1%)	172 (20.5%)	0
Night 2 (558 epochs)	40 (7.2%)	71 (12.7%)	0
Nap 1 (113 epochs)	36 (31.9%)	15 (13.3%)	0
Nap 2 (233 epochs)	31 (13.3%)	35 (15.0%)	0

Why “100% coverage” can be misunderstood

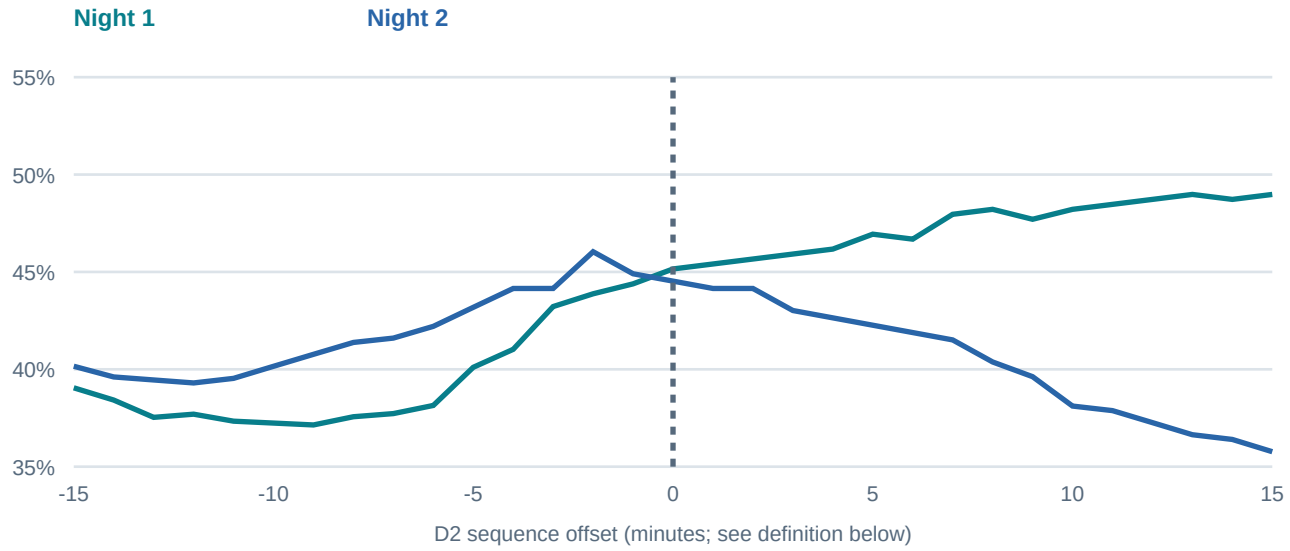
All four saved H10 reports show 100% coverage because coverage accepts an epoch if activity **or** HR is present. Activity is available in every epoch, even where HR or breathing is null. Thus 100% coverage does not imply 100% cardiac or respiratory availability, nor equal confidence in every stage. Counts above cover the whole stored feature array, including boundaries.

Breathing values are accepted proxies rather than direct airflow measurements. D2 overnight mean SpO2 values of 98% and 97% are summary means with no nadir here; they cannot exclude sleep-disordered breathing. No RDI result or reference apnea assessment is available.

Sources: polarCopy.report feature arrays; watchCopy.summary; [S1]. Missing feature epochs can reflect rejection criteria or insufficient data, not necessarily a physical disconnection.

16 Are the clocks misaligned?

Sensitivity analysis: shifting the D2 sequence does not reconcile the stage patterns.



Night	Original stage match	Highest in +/-15 min scan	Offset at that maximum
Night 1	45.15%	48.98%	+13 and +15 min
Night 2	44.53%	46.04%	-2 min

What the offset means

At offset k, H10 minute i is compared with D2 minute i+k. A positive value compares H10 with a later D2 label, equivalent to moving the D2 plotted sequence earlier. Minutes without data after the shift are excluded. The plot is an exploratory stress test, not a clock correction.

A simple global clock problem would be more convincing if a similar small offset substantially improved both nights. Here, the improvement is limited and the preferred direction differs. Night 1's maximum is also near the search boundary. These observations make a single fixed clock offset an inadequate explanation for the poor stage matching.

Residual alignment limitations

The app bins H10 epochs by their start timestamps and takes the first stage in a tied minute. It does not weight how many seconds an epoch overlaps a clock minute. D2 supplies whole-minute labels. Arrival-time anchoring and reconnection handling can add timing uncertainty, and no external hardware clock reference is supplied. These limitations matter for transitions but do not turn the observed results into proof of a clock error.

Additional analysis from eligible saved arrays, offsets -15 through +15 minutes, with original results retained elsewhere. Selecting an offset to maximize agreement on the same data would overfit; no adjusted accuracy claim is made.

17 App findings that affect interpretation

The software audit identifies several presentation and inference risks.

Finding	Evidence and consequence
Generic nap sleep shown as light	src/d2/night.ts includes code 5 in lightMin; syncsleep.ts correctly keeps it unstaged for agreement. Label the nap total “unstaged sleep” consistently.
Zeros create false warnings	Nap avgHR, avgHrv and breathing are zero. buildReports subtracts them as numbers, producing false HR/HRV gaps and unnecessary watch-fit advice.
Coverage is too broad	Activity OR HR defines coverage. A separate valid-HR, valid-respiration and complete-feature coverage would show uncertainty.
Awakenings differ by rule	H10 uses >=2-minute internal runs; the D2 fallback counts each internal awake run. Combined counts are rounded device midpoints.
Nap score fields persist	Stored H10 nap scores 40/47 exist although the changelog and UI policy suppress nap scoring. They should not be interpreted as poor nap quality.

Combined is a third rule set, not adjudicated truth

The fusion algorithm favors deep/REM over light, follows the H10 when deep and REM conflict, and treats brief awake/asleep disagreements as sleep; a discordant run of at least 5 minutes becomes awake. Generic D2 asleep inherits the other device's stage. These are explicit choices, not an independently validated consensus. [S2]

For Night 1, the combined report assigns 93 deep minutes and 123 REM minutes, compared with H10's 75.5/76 and D2's 22/90. Fusion can therefore exceed both devices' totals by taking differently timed episodes from each. Minute conversion also changes precision. Treat the fused architecture as a derived visualization, not a more accurate measurement.

The combined awakening count is a rounded midpoint, not the number of awake runs in the combined hypnogram. A larger combined stage total or a better-looking score must not be used to decide which device was correct.

Sources: [S2] combine, figuresOf, midWakes, suggestions/buildReports; [S3] figures; [S4] changelog. No application source files were modified for this report.

18 Practical next steps

A stronger personal comparison and targeted software improvements.

For the next recordings

Collect at least 7-14 additional paired nights if practical, alongside separately identified naps. Keep device placement and sleep settings consistent, retain firmware/app versions and note whether automatic blood-pressure measurements or other disturbances occur. Start H10 before lights out and end after getting up; note the actual times in a short diary.

Record lights out, remembered awakenings, final waking, getting up and whether you were lying still awake. A diary does not replace EEG, but helps identify quiet-wake errors and recordings that capture only part of the night. Keep the phone within Bluetooth range and check signal availability before sleeping.

For the app: highest-value changes

Priority	Recommended change
1	Represent code 5 as unstaged sleep in every total, chart, export and combined explanation; never silently equate it with N1/N2.
2	Normalize physiological zero placeholders to unavailable using field-specific validity checks. Suppress differences and automated advice when either value is missing.
3	Expose per-feature validity, not just activity-or-HR coverage. Flag stage estimates made with missing cardiac or breathing inputs.
4	Display shared-window totals alongside full-recording totals; document minute rounding and stage ties.
5	Use a common awakening definition for an optional comparison view, while preserving the original device summaries.
6	Keep fusion visibly separate from device outputs and retain method version, source snapshots and report-generation timestamps.

If the goal is clinical accuracy

The useful next test is simultaneous reference sleep scoring, not choosing whichever wearable reports more deep sleep. Clinical evaluation is appropriate for persistent daytime sleepiness, witnessed pauses in breathing, troublesome insomnia or other concerning symptoms. A clinician can decide whether a sleep study is warranted; this export alone cannot diagnose or exclude a sleep disorder. [5]

The suggested collection period is a practical protocol for trend assessment, not a claim that 7-14 nights establish clinical validation.

19 Conclusions and evidence trail

What can be concluded from this backup, and how the analysis can be checked.

Conclusion: the paired systems broadly agree on sleep duration, but their overnight stage labels are not interchangeable. H10/OpenBeat and D2 naps support only a duration and binary-sleep comparison because the D2 nap data are unstaged. Neither device is a clinical reference.

Reproducibility notes

The four current eligible pairs were selected before numerical analysis. Saved report histories were not counted as extra sessions. Original minute arrays were reconstructed from embedded stages and timestamps; agreement statistics were recomputed exactly. Lag results are exploratory, use only the nights and do not replace the original clock alignment. Raw H10 signals were not reclassified in this analysis.

Backup: openbeat-backup-2026-11-06-10-57.json, exported 2026-11-06T10:57:50.956Z; appVersion 1.32.1 (5 Nov 2026). SHA-256: bc38d8055ffe5bf324c814479673407f7d1fae3e8789b1d6ed94a47323519b91

Supplied implementation sources

[S1] src/sleep.ts - feature extraction, sleep/wake, stage scores, smoothing and sleep score.
[S2] src/syncsleep.ts - matching, alignment, fusion, agreement and suggestions.
[S3] src/d2/night.ts - code-5 totals, trim boundaries and awakening fallback.
[S4] CHANGELOG.md - method 2 / v1.29.0 and nap-scoring policy; package.json - v1.32.1.
Also inspected: src/summary.ts (saved report format), src/syncsleepui.ts (display/difference handling), src/d2/sleepfile.ts (imported fields).

Source root: /Users/mike/Desktop/01 Elitebook 13 Apr 2026/Documents/programming/ECG Analysis/GitHub uploads/2026-11-06 0001 publish/

External primary references - accessed 6 November 2026

[1] Polar H10 SDK features

[2] Huawei WATCH D2 sensor specifications

[3] Huawei sleep-monitoring principles

[4] NHLBI: Sleep phases and stages

[5] AASM technology evaluation update (2021), doi:10.5664/jcsm.9580

[6] Huawei developer sleep-status enumeration

Huawei documentation describes product-family principles; the backup does not establish the watch's TruSleep version. Clinical context is explanatory and does not supply a reference label for these recordings.